1. **If a model suffering from overfitting, which techniques would you prioritize to reduce it, and why?**

- **Get More Data (Increase Training Data):** Gather more information to help the model better grasp the underlying patterns in the data.
- **Regularization:** Regularization techniques add penalty terms to the loss function to prevent the model from fitting the training data too closely.
- **Data Augmentation:** Data augmentation is the process of creating new samples by applying random transformations to your training data.
- **Dropout:** Avoid over-reliance on specific neurons.
- **Reduce Model Complexity:** To avoid overfitting, select simpler model architecture.

2. **Prepare report , briefly compare these four optimizers: Standard GD, SGD, Momentum, RMSprop, and Adam.**

- **Standard GD**
    a. **The Idea:** It finds the optimal values of weights and biases by gradually reducing the error between predicted and actual outputs (it uses entire dataset to compute gradient.)
    b. **Speed:** very slow.
    c. **Tuning:** Needs careful tuning. (Hard)
- **SGD**
    a. **The Idea:** It uses one sample to compute gradient at a time.
    b. **Speed:** Faster than Standard GD.
    c. **Tuning:** Needs careful tuning. (Hard)
- **Momentum**
    a. **The Idea:** It adds velocity (memory of past gradients).
    b. **Speed:** Faster training.
    c. **Tuning:** Needs careful tuning. (Medium)
- **RMSprop**
    a. **The Idea:** It adapts the learning rate for each parameter individually by considering the magnitude of recent gradients for those parameters.
    b. **Speed:** Fast.
    c. **Tuning:** Lower Learning Rate Sensitivity. (Easy)

- **Adam**
    a. **The Idea:** combines the advantages of Momentum and RMSprop techniques to adjust learning rates during training.
    b. **Speed:** Fastest.
    c. **Tuning:** Easiest + smartest.

- The fastest optimizer to reach the lowest error is **Adam**.
- Optimizer requires a lot of trial-and-error to find the right Learning Rate is **Standard GD** and **SGD.**
- The smartest optimizer is **Adam**.
- My choice for massive AI project is **Adam** because it is the fastest and smartest optimizer.

3. Why do modern deep learning models often use Adam optimizer together with dropout and weight initialization strategies?

because each technique solves a different challenge during training.

- **Adam optimizer:** It provides fast and stable convergence through adaptive learning rates.
- **Dropout:** It helps reduce overfitting by disable random neurons during training.
- **Weight initialization:** ensures that training starts properly by preventing vanishing and exploding gradients.

4. A deep neural network shows the following behavior:
Training error: 1% , Dev error: 13%

1) It is High Variance problem (Overfitting),
   the first three methods (Increase training data, add L2 regularization, use dropout) can solve the problem.

2) Weight Initialization addresses training stability.
3) It helps the model to learn the underlying patterns instead of memorizing noise.